Course Code	21CSC558J	Course Name	DEEP LEARNING APPROACHES	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	0	2	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Data Science and Business Systems*	Data Book / Codes / Standards	*Nil*		

Course Learning Rationale (CLR):	***The purpose of learning this course is to:***
CLR-1:	*Understand the foundations of neural network concepts*
CLR-2:	*Acquire various optimization, regularization and normalization approaches in neural networks*
CLR-3:	*Appreciate the architectures of Convolutional neural network and Recurrent Neural Networks*
CLR-4:	*Comprehend the functionalities of Auto encoders, transformers and Generative networks*
CLR-5:	*Explore the application of deep neural network architectures in language, text, speech, video and image processing*

Course Outcomes (CO):	***At the end of this course, learners will be able to:***	***Programme Outcomes (PO)***		
		1	***2***	***3***
CO-1:	*Learn the fundamental concepts of neural networks*	3	-	-
CO-2:	*Recognize the significance of optimization, regularization and normalization techniques in neural networks*	-	3	-
CO-3:	*Implement the Convolutional neural network and Recurrent Neural Networks*	2	-	1
CO-4:	*Learn about the Auto encoders, transformers and Generative networks*	-	1	3
CO-5:	*Apply deep neural network architectures for language, text, speech, video and image processing*	-	2	2

Module-1 - Fundamentals of Deep Learning **15 hour**

Introduction to neural networks-machine learning vs deep learning, biological motivation of neural networks, fundamentals of tensor flow-data structures in tensor flow, Perceptron model-perceptron learning rule, learning Ex-OR problem, activation functions- role of activation function, different types of activation functions, feedforward neural networks, back propagation in neural networks-chain rule, loss function s- binary cross entropy loss, categorical cross entropy loss

Lab 1: exploring the environment of tensor flow

Lab 2: Building Programs to Perform Basic Operations in Tensors, regression

Module-2 - Optimization, Regularization and Normalization in Deep Neural Networks **15 Hour**

Gradient descent algorithm-types of gradient descent, RMS prop, Adagrad, ADAM optimiser, Unit Saturation- Vanishing And Exploding Gradient, Underfitting, Overfitting And Learning Rate, Regularization Techniques In Neural Networks-LASSO Regression, Ridge Regression, Dropouts, Early Stopping, Normalization techniques in Neural Networks- Batch Normalization, Group Normalization, Instance Normalization

Lab 3: Building neural network using Keras

Lab 4: Building programs to optimize the neural network using gradient descent. normalization techniques

Module-3 - Convolutional Neural Networks and Recurrent Neural Networks **15 Hour**

Convolutional neural networks- biological motivation, convolution operation, Pooling layer, types of pooling, Fully connected layer, Applications in Computer vision, Imagenet, Transfer Learning, Sequence modelling- Recurrent Neural Networks- RNN Topologies, Difficulty in training in RNN, Long Short-Term Memory networks, Bi-Directional LSTM, Gated, Recurrent units.

Lab 5: Building Programs for Compute vision problems using CNN

Lab 6: Building Programs for multi class computer vision problem using CNN, LSTM

Module-4 – Autoencoders, Transformers and Generative Networks **15 Hour**

Auto Encoders and Decoders- introduction using auto encoders, educational auto encoders, regularized auto encoder, variational Auto encoders, denoising autoencoder, applications of auto encoder, Generative Adverserial Networks (GAN), Transformers-attention mechanisms BERT.

Lab 7: Building Programs to create Autoencoders with Keras

Lab 8: Building Programs to create a denoising autoencoders, GAN

Module-5 - Deep Architectures for Heterogeneous Data Processing **15 Hour**

GPT, auto regressive models, stability diffusion models, Vision and Language applications-Image captioning, Visual QA, Visual Dialog, Pixel RNNs, Cycle GANs, Progressive GAN, Stcak GAN, Pix2Pix

Lab 9: implement stable diffusion model

Lab 10: Implement Image caption generation models, pre trained model

Learning Resources	*1. Chollet, F, "Deep learning with Python", Manning Publication, 2019* *2. Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", MIT Press, 2017.* *3. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.*	*4. J ason Brownlee, "Deep Learning with Python",ebook,2016* *5. Seth Weidman Seth Weidman, "Deep Learning from Scratch: Building with Python from First Principles",Oreilly, 2019.* *6. Antonio Gulli, "TensorFlow 1.x Deep Learning Cookbook," Packt Publishing Limited, 2017.*

Learning Assessment

		Continuous Learning Assessment (CLA)					
	Bloom's Level of Thinking	Formative CLA-1 Average of unit test (45%)		Life Long Learning CLA-2- Practice (15%)		Summative Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	-	15%	15%	-
Level 2	Understand	25%	-	-	25%	25%	-
Level 3	Apply	30%	-	-	30%	30%	-
Level 4	Analyze	30%	-	-	30%	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers

Experts from Industry	**Experts from Higher Technical Institutions**	**Internal Experts**
1.Dr.Sagarika, Senior Data Scientist, Great Learning	*1.Dr. Anandhakumar P* *Professor, Madras Institute of Technology, Chrompet*	*1. Dr M Prakash, SRMIST* *2. Dr. S. Sharanya, SRMIST*